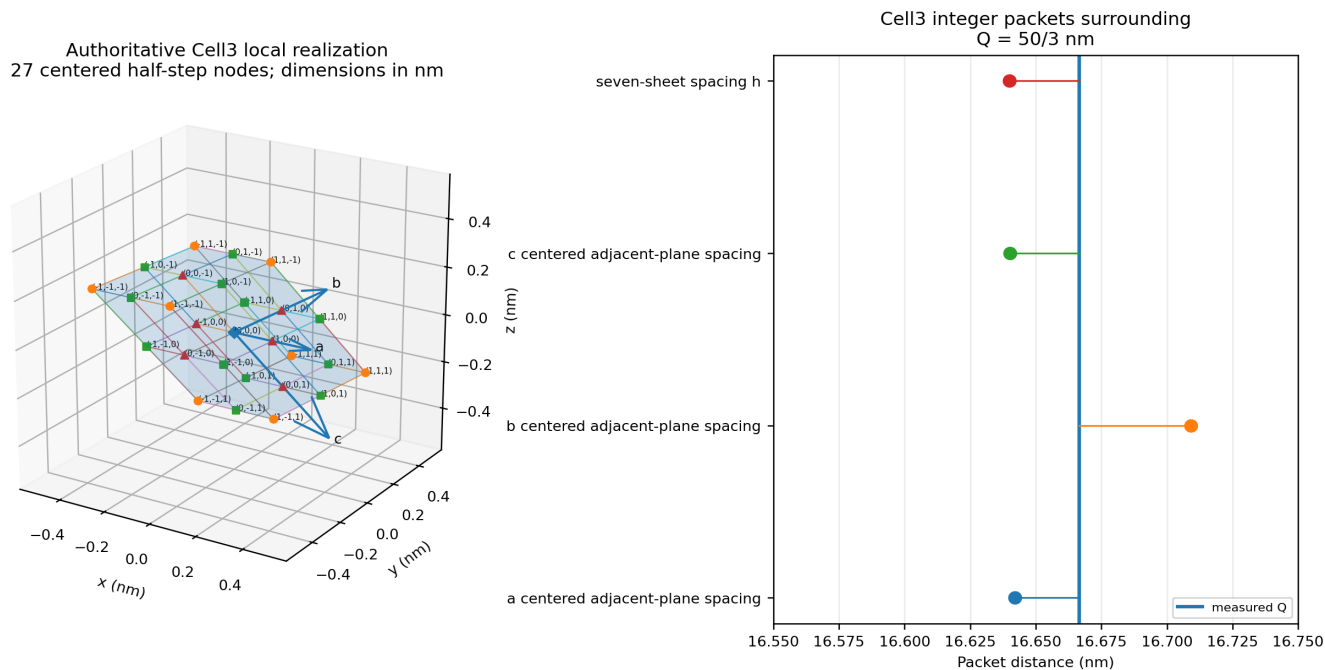


K-FIELD

CELL3 VERSUS SPECTROPHOTOMETER QUANTIZED INTERPLANAR DISTANCE LADDER

Integer Cell3 plane packets, cross-node spacings, and the geometric interpretation of the measured $Q = 50/3$ nm ladder



Cover figure. Left: the authoritative oblique Cell3 local realization with all 27 centered half-step nodes, nearest-neighbor links, address labels, and ordered basis vectors. Right: the measured principal interval Q compared directly with the nearest integer packets of the three centered coordinate-plane spacings and the seven-sheet spacing. Every plotted value is generated from the exact numerical ledgers used by this paper.

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This paper is a standalone educational and archival technical reference. All numerical tables and figures are generated from the measured spectrophotometer ladder ledgers, the authoritative Cell3 basis, and exact arithmetic transformations of those inputs. No synthetic measurement series is introduced.

Central result: the spectrophotometer principal interval Q is not a primitive Cell3 edge. It is a supercell packet that closes simultaneously against integer Cell3 plane counts and an especially direct (1,1,-1) cross-node count.

Document item	Normative value
Measured carrier	T = 54.17 s
Measured principal interval	Q = 50/3 nm = 16.6666667 nm
Measured quarter increment	q_d = Q/4 = 25/6 nm = 4.1666667 nm
Cell3 scaffold	27 centered half-step nodes p_L = (1/2) B L, L in {-1,0,1}^3
Direct cross-node lock	Q approximately 50 r_11-1 approximately 25 D_11-1
Coordinate-plane packet	approximately (101, 74, 55) centered adjacent planes
Seven-sheet packet	approximately 110 sheet intervals
Machine-readable companion	K-Field_cell3_vs_spectrophotometer_quantized_interplanar_distance_ladder_1785507240.json

Abstract

This archival reference compares the raw spectrophotometer interplanar-distance ladder with the authoritative K-Field Cell3 geometry. The measured ladder has principal interval $Q = 50/3 \text{ nm} = 16.6666667 \text{ nm}$ and quarter increment $q_d = 25/6 \text{ nm} = 4.1666667 \text{ nm}$. The Cell3 root object is the ordered oblique basis $B = [a \ b \ c]$ in nanometres. Its centered 27-node scaffold is $p_L = (1/2) B L$ for integer addresses L in $\{-1,0,1\}^3$. The coordinate-plane families have centered adjacent spacings $d_a/2 = 0.164773264855 \text{ nm}$, $d_b/2 = 0.225798689862 \text{ nm}$, and $d_c/2 = 0.302548870757 \text{ nm}$. The seven-sheet family has adjacent spacing $h = 0.151272688042 \text{ nm}$. The Cell3 body-diagonal address $(1,1,-1)$ has center-to-node radius $r_{11-1} = 0.333198543971 \text{ nm}$ and antipodal span $D_{11-1} = 0.666397087941 \text{ nm}$. Direct comparison gives $Q/r_{11-1} = 50.0202266$ and $Q/D_{11-1} = 25.0101133$, so one measured Q interval is approximately 50 center-to-node radii or 25 complete antipodal cross-node spans. The same Q interval is approximated by 101 a-channel centered planes, 74 b-channel centered planes, 55 c-channel centered planes, and 110 seven-sheet intervals. The recurrent spectrophotometer plane tracks therefore admit two simultaneous integer descriptions: quarter- Q ladder addresses and Cell3 supercell counts. The complete paper supplies the exact basis, metric, all 27 nodes, six boundary planes, seven sheets, cross-node geometry, packet residuals, recurrent-track mappings, and a machine-readable reconstruction package.

Keywords: K-Field; Cell3; AS7341 spectrophotometer; quantized interplanar distance; reciprocal plane spacing; cross-node spacing; supercell packet; 27-node lattice.

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1. Complete Result Statement

The strongest raw geometric interpretation is that the spectrophotometer ladder represents integer supercell packets of the canonical Cell3 plane and cross-node spacings, rather than a new primitive Cell3 dimension. The measured ladder is:

$$Q = 50/3 \text{ nm} = 16.6666667 \text{ nm}; \quad q_d = Q/4 = 25/6 \text{ nm} = 4.1666667 \text{ nm}$$

The recurrent spacing states occupy 17.25Q, 18Q, 18.25Q, 19.75Q, 20Q, 20.5Q, 21Q, 25Q, and 28Q. The canonical Cell3 is the oblique 27-node half-step lattice $p_L = (1/2) B L$, where L belongs to $\{-1, 0, 1\}^3$.

The Cell3 body-diagonal node address (1,1,-1) has center-to-node spacing $r_{11-1} = (1/2)|a+b-c| = 0.333198543971 \text{ nm}$. Its antipodal cross-node span is $D_{11-1} = 2 r_{11-1} = 0.666397087941 \text{ nm}$.

$$Q/r_{11-1} = 50.0202266; \quad Q/D_{11-1} = 25.0101133$$

Therefore Q approximately equals 50 r_{11-1} and 25 D_{11-1} . Numerically, 50 $r_{11-1} = 16.6599272 \text{ nm}$. The displacement from Q is 0.00673947 nm, or 0.0404 percent of Q .

This is the cleanest direct Cell3 correspondence: one observed Q interval is approximately 25 complete antipodal (1,1,-1) cross-node spans.

The canonical primitive plane spacings are $d_a = 0.329546529709 \text{ nm}$, $d_b = 0.451597379724 \text{ nm}$, and $d_c = 0.605097741515 \text{ nm}$. The associated adjacent centered half-step planes are separated by $d_a/2$, $d_b/2$, and $d_c/2$. The same observed Q interval lies near the integer plane-count packet $(N_a, N_b, N_c, N_h) = (101, 74, 55, 110)$.

In this form Q can be read simultaneously as approximately 101 adjacent a-planes, 74 adjacent b-planes, 55 adjacent c-planes, and 110 K-Field sheet intervals. The closest primitive-edge relations are Q approximately 50 $|a|$ and Q approximately 21 $|c|$.

The recurrent solutions map to integer cross-node counts. Whole- Q states are especially transparent: 18Q maps to approximately 900 center-to-node units, 20Q to 1000, 21Q to 1050, 25Q to 1250, and 28Q to 1400 before the small Q -to-Cell3 scale displacement is applied.

The results resolve into two related structures: Q approximately 25 antipodal (1,1,-1) cross-node spans, and Q approximately (101,74,55) adjacent coordinate-plane spacings with an accompanying 110h sheet packet. The quarter ladder increment q_d is therefore a subdivision of a Cell3 supercell period. It is not the primitive Cell3 edge length.

The observed interplanar distances therefore approximate integer Cell3 plane-crossing counts and integer cross-node traversal counts. The strongest single correspondence is the (1,1,-1) cross-node direction, where the principal observed interval closes at approximately 50 center-to-node radii or 25 antipodal node spans.

2. Measurement Basis and Quantized Ladder

The source record contains 1,000 valid spectrophotometer frames. The cadence-resolved analysis used 270-sample windows, equal to three nominal 90-step turns, advanced by a 90-sample cadence. Nine complete windows were available. The fixed $T = 54.17$ s component supplied phase angles for the two-wavelength Bragg inversion.

The resulting plane-group distances were indexed by q_d . For any raw group distance d_g , the ladder address is $k = \text{round}(d_g/q_d)$, the representative distance is $d_hat_g = k \cdot q_d$, and the residual is $\epsilon_g = d_g - d_hat_g$.

$$k = \text{round}(d_g/q_d); \quad d_hat_g = k \cdot q_d; \quad \epsilon_g = d_g - d_hat_g$$

The 41 cadence-resolved plane-group centers occupy 30 distinct quarter indices. The nine recurrent tracks occupy the nine Q levels retained throughout this paper. This report does not replace those raw distances; it overlays them with the authoritative Cell3 metric to expose integer supercell addresses.

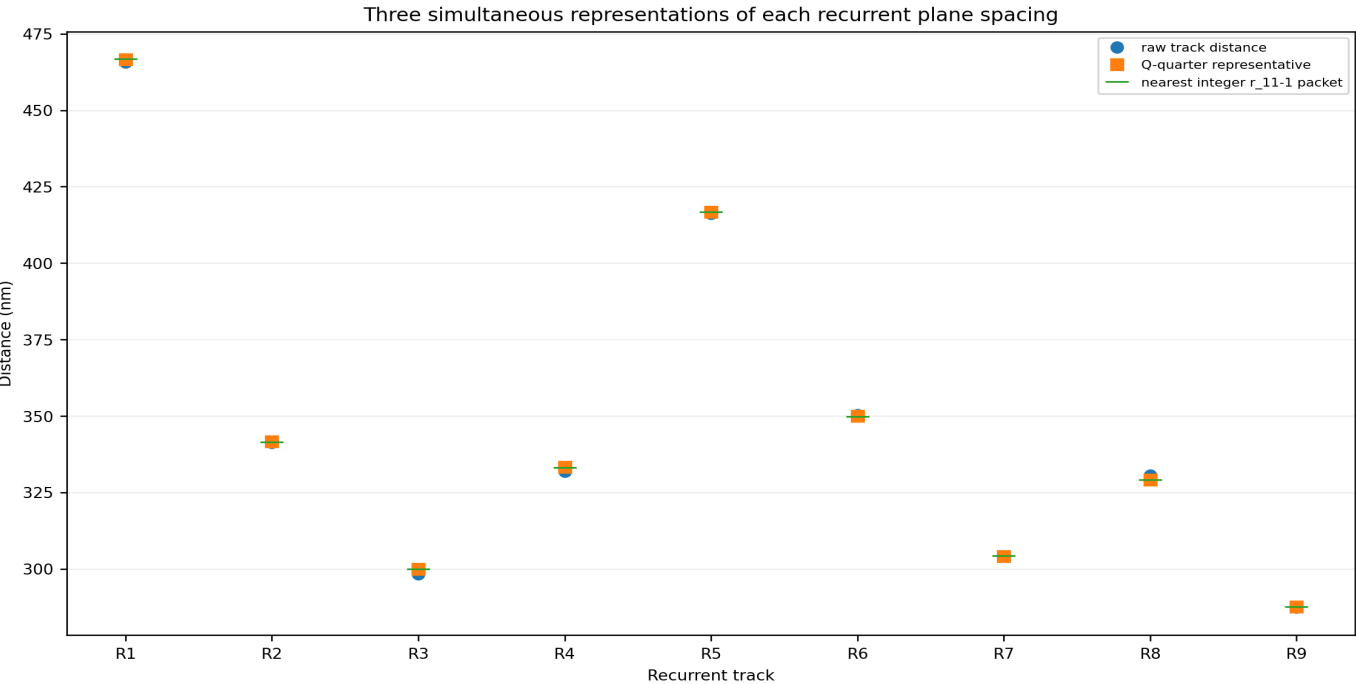


Figure 1. For each recurrent track, the raw Bragg distance, its nearest quarter-Q representative, and its nearest integer (1,1,-1) cross-node-radius packet are shown together. The three representations preserve raw measurement, measured ladder state, and Cell3 integer address.

Quantity	Value
Valid frames	1000
Sliding window	270 samples
Cadence	90 samples
Complete windows	9
Plane-group centers	41
Recurrent tracks	9
Principal interval Q	16.666666666667 nm
Quarter increment q_d	4.166666666667 nm

3. Authoritative Cell3 Geometry

The Cell3 root object is the ordered basis $B = [a \ b \ c]$. Its ICRS columns carry absolute orientation. An equivalent local engineering frame preserves all lengths, angles, faces, volume magnitude, reciprocal spacings, and handedness while placing a on the local x -axis and b in the local x - y plane.

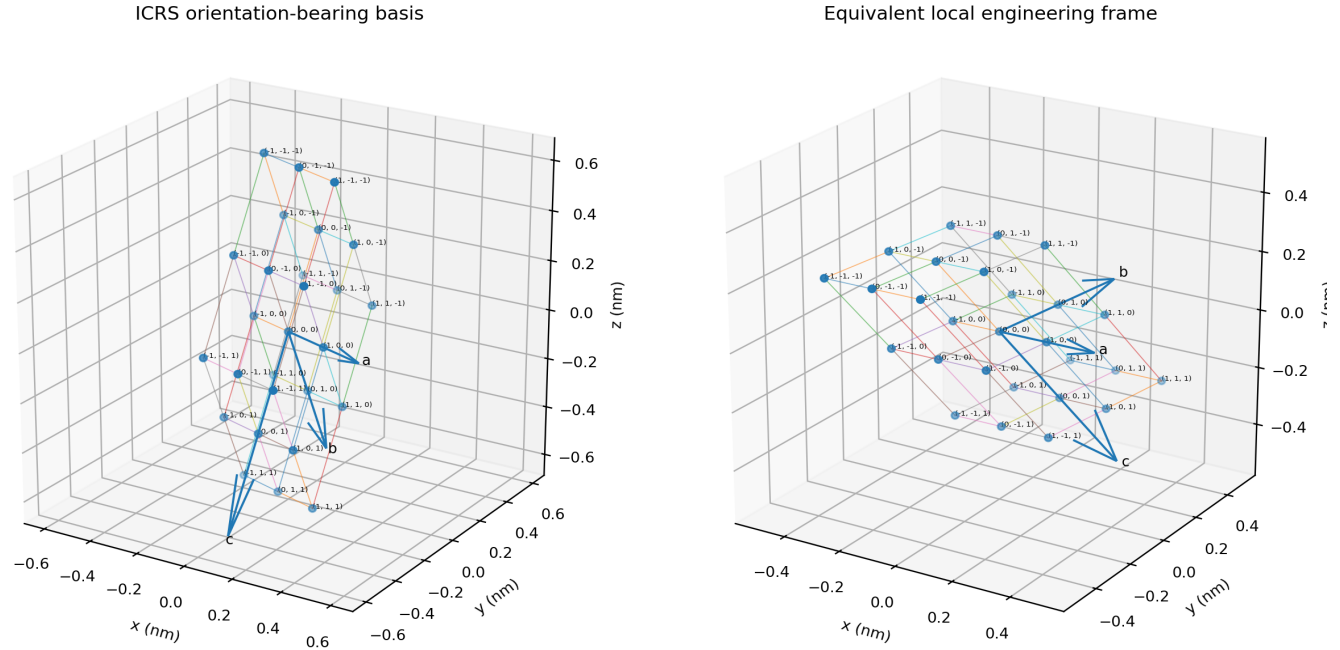


Figure 2. The same Cell3 intrinsic geometry in two coordinate frames. Left: the authoritative ICRS orientation-bearing basis. Right: the local engineering realization used for readable dimensioned figures. All 27 centered half-step nodes are shown and labeled in both frames.

Vector	ICRS x	ICRS y	ICRS z	Local x	Local y	Local z	Norm (nm)
a	0.327501242070	-0.054897007886	-0.037487502896	0.334179679073	0.000000000000	0.000000000000	0.334179679073
b	0.035330775844	0.207319850441	-0.547401085392	0.061973646450	0.583126915468	0.000000000000	0.586410890415
c	-0.010129317627	-0.389787702364	-0.689734408363	0.131477878067	0.494308507778	-0.605097741515	0.792319765038

The direct metric $M = B^T B$ stores squared lengths and pairwise dot products. The reciprocal metric $R = M^{-1}$ stores the dual plane geometry. The cell volume is $V = |\det B|$, and the opposite-face rule gives $d_a = V/A_{bc}$, $d_b = V/A_{ac}$, and $d_c = V/A_{ab}$.

$$V = 0.117914891916764 \text{ nm}^3; A_{ab} = 0.194869165469925 \text{ nm}^2; A_{ac} = 0.261106235799411 \text{ nm}^2; A_{bc} = 0.357809539128890 \text{ nm}^2$$

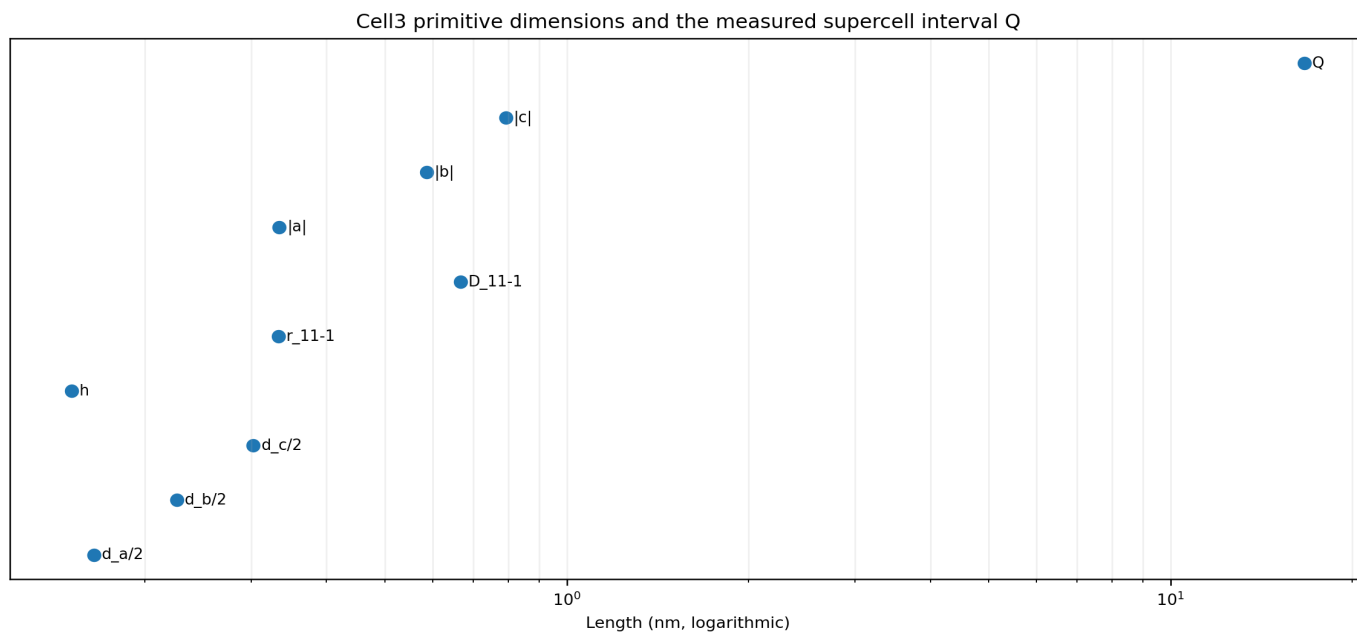


Figure 3. Logarithmic scale hierarchy from primitive Cell3 sheet and plane spacings through edge and cross-node dimensions to the measured supercell interval Q.

4. The 27-Node Centered Scaffold

Each node has integer address $L = (i,j,k)$ with i,j,k in $\{-1,0,1\}$. Physical position is $p_L = (1/2)(i a + j b + k c)$. The half-step construction gives one body center, six face centers, twelve edge centers, and eight corners.

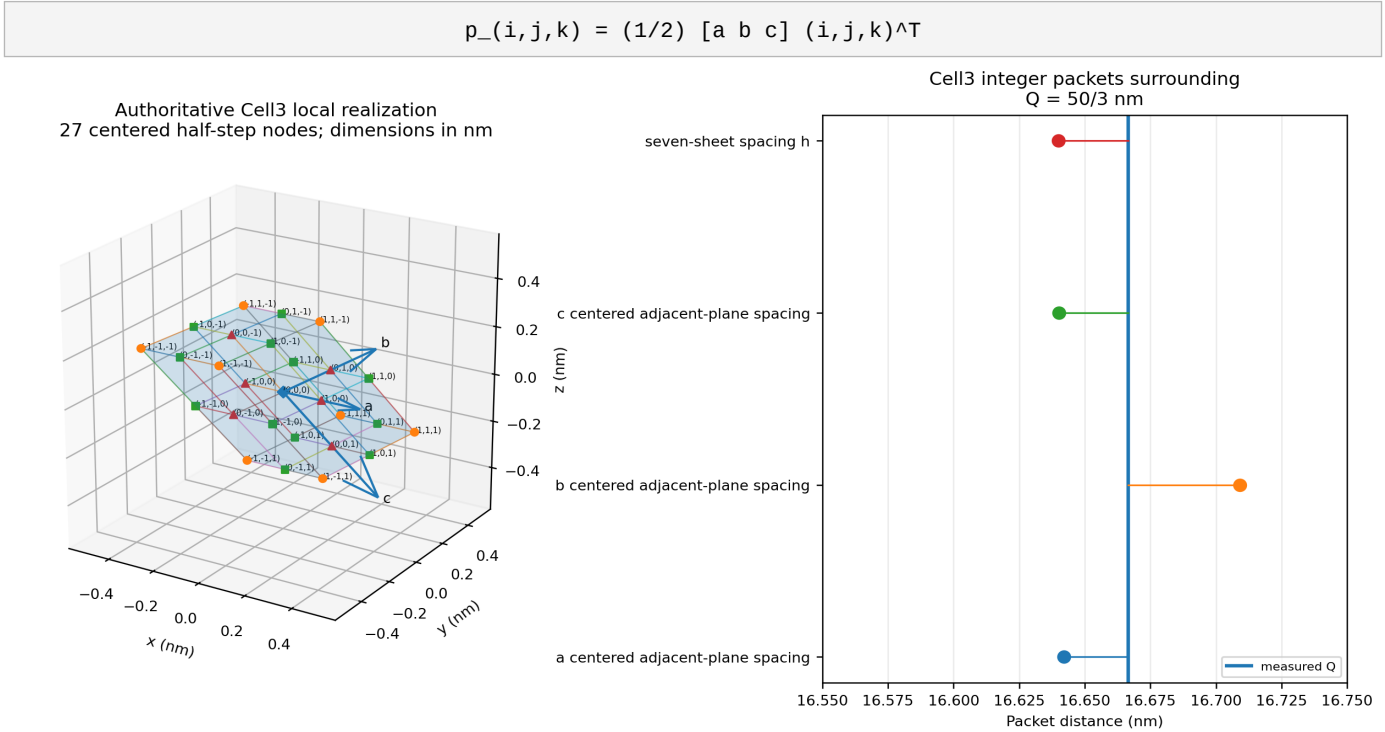


Figure 4. Left panel: the complete dimensionally correct 27-node local Cell3. Address labels, nearest-neighbor half-step links, boundary faces, and basis vectors are generated directly from the node equation.

The 27-node cloud is not an illustrative cubic surrogate. Its oblique shape is fixed by the authoritative basis. A nearest-neighbor address step along i , j , or k is a physical half-vector $a/2$, $b/2$, or $c/2$. The boundary corners occur at all sign combinations of $(a+b+c)/2$.

Node class	Address rule	Count
Body center	three zero coordinates	1
Face center	two zero coordinates	6
Edge center	one zero coordinate	12
Corner	no zero coordinates	8

5. Coordinate-Plane and Sheet Families

Fixing one address coordinate generates a nine-node plane. For the a-channel family, $i = -1, 0, \text{ or } +1$; for the b-channel family, j is fixed; for the c-channel family, k is fixed. Because node positions use half steps, adjacent centered planes are separated by $d_a/2$, $d_b/2$, and $d_c/2$.

a-channel planes: adjacent spacing 0.164773264855 nm

b-channel planes: adjacent spacing 0.225798689862 nm

c-channel planes: adjacent spacing 0.302548870757 nm

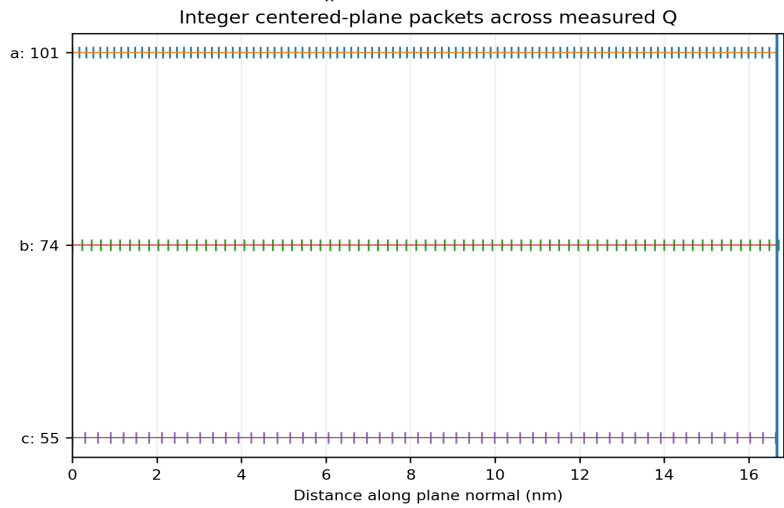
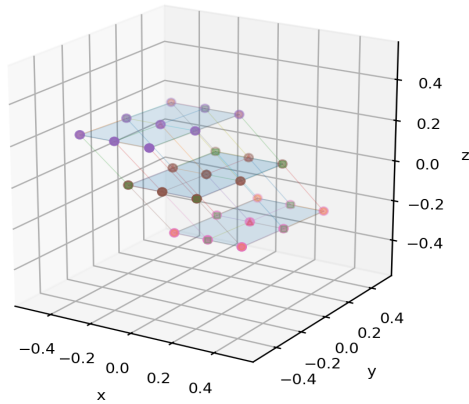
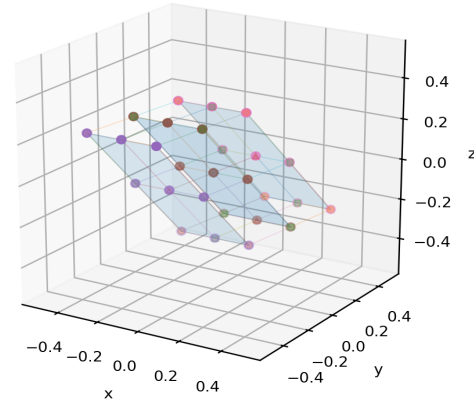
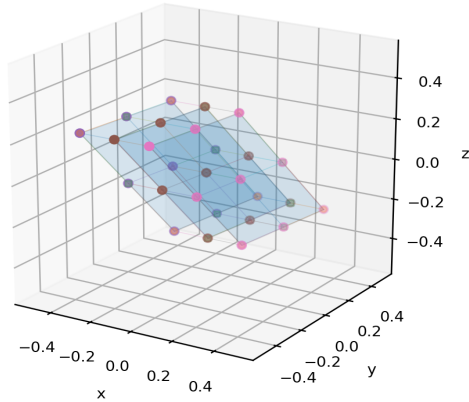


Figure 5. Top and lower-left: the three coordinate-plane families rendered in the exact oblique Cell3. Each family contains the -1, 0, and +1 address planes and all nine nodes on each plane. Lower-right: the integer packet of each centered plane spacing across one measured Q interval.

Family	Primitive spacing d (nm)	Centered adjacent spacing $d/2$ (nm)	Unit normal in local frame	Nodes per address plane
a	0.329546529709	0.164773264855	$(-0.986135754, +0.104804678, -0.128655566)$	9
b	0.451597379725	0.225798689862	$(+0.000000000, -0.774440980, -0.632646164)$	9
c	0.605097741515	0.302548870757	$(+0.000000000, +0.000000000, +1.000000000)$	9

The seven-sheet family is obtained from sheet address $s = i+j+k$. The sheet normal N_{sheet} is the normalized solution of $B^T N$ proportional to $(1,1,1)$. Every node on the same integer s has the same physical sheet coordinate $N_{\text{sheet}} \cdot p = s h$.

Seven Cell3 sheets; adjacent spacing $h=0.151272688042$ nm

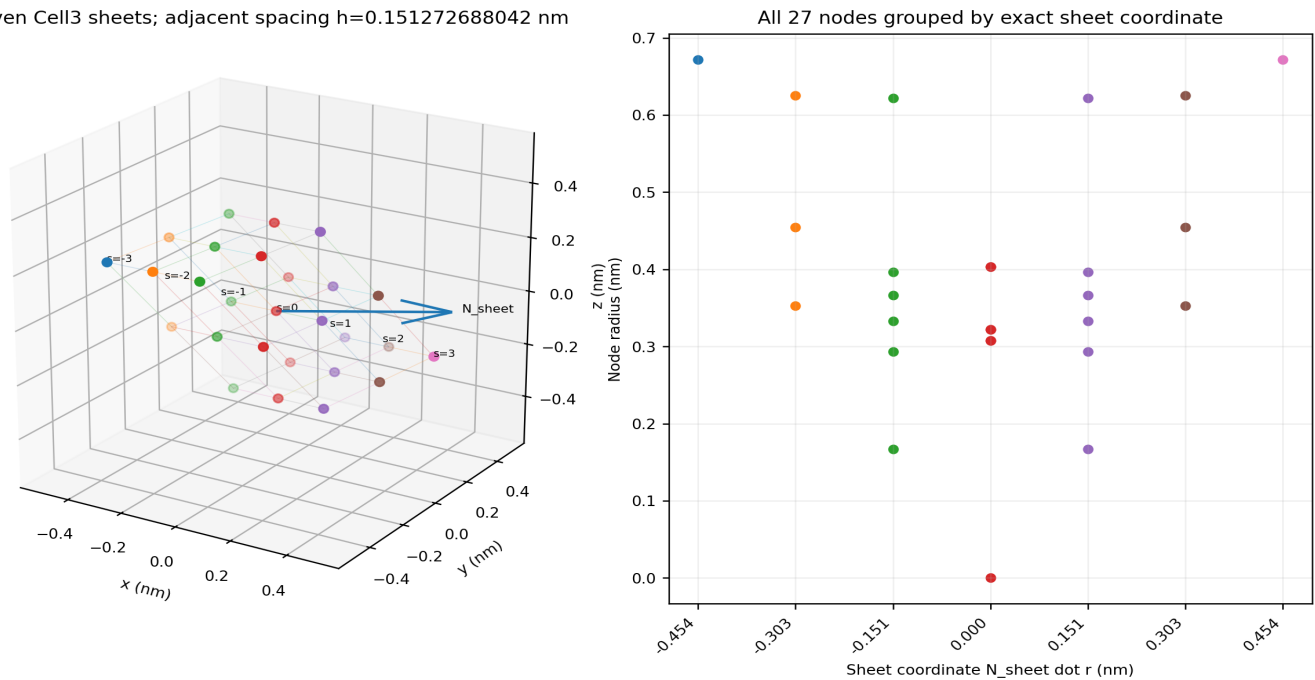


Figure 6. Left: all seven sheets $s=-3$ through $s=3$ in the dimensionally correct Cell3, with every node assigned to its exact sheet. Right: the 27 nodes grouped by exact sheet coordinate and radial distance. Adjacent sheet spacing is $h = 0.151272688042$ nm.

6. The (1,1,-1) Cross-Node Geometry

The node address (1,1,-1) identifies the point $p = (a+b-c)/2$. Its antipode is $-p$. The line through the two nodes passes through the Cell3 body center and is one of the four oblique cross-body diagonal directions.

$$r_{11-1} = (1/2) |a+b-c|; \quad D_{11-1} = |a+b-c| = 2 r_{11-1}$$

Cell3 (1,1,-1) cross-node line
 $r=0.333198543971$ nm; $D=0.666397087942$ nm

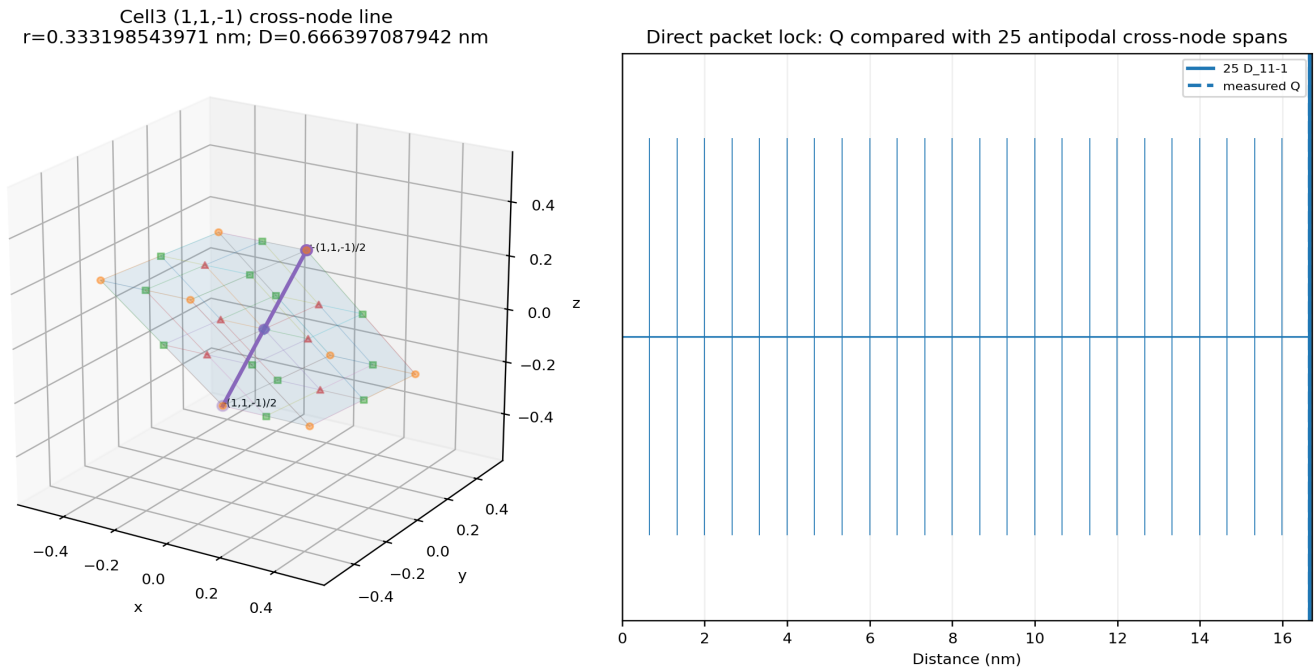


Figure 7. Left: the complete cross-node line between the antipodal (1,1,-1) nodes inside the authoritative local Cell3. Right: 25 exact antipodal spans laid end-to-end against the measured Q endpoint.

Cross-node quantity	Value
Address	(1,1,-1)
Local vector a+b-c	(0.264675447456, 0.088818407690, 0.605097741515) nm
Center-to-node radius r ₁₁₋₁	0.333198543970849 nm
Antipodal span D ₁₁₋₁	0.666397087941697 nm
Q/r ₁₁₋₁	50.020226583358
Q/D ₁₁₋₁	25.010113291679

7. Direct Cross-Node Closure of Q

Multiplying the Cell3 center-to-node radius by 50 gives the same packet as multiplying the antipodal span by 25, because $D_{11-1} = 2 r_{11-1}$.

$$50 r_{11-1} = 25 D_{11-1} = 16.659927198542437 \text{ nm}$$

The measured interval is $Q = 16.666666666666668 \text{ nm}$. The raw endpoint displacement is $Q - 50 r_{11-1} = 0.006739468124231 \text{ nm}$. Relative to Q, the displacement is 0.040436809 percent.

The number 50 is therefore not introduced as a free fit parameter after the fact. It is the nearest integer cross-node count generated by dividing the measured Q by the authoritative Cell3 radius. The equivalent antipodal count is 25.

Q approximately 50 r₁₁₋₁ approximately 25 D₁₁₋₁

This correspondence is structurally stronger than a comparison to a single edge because the cross-node object simultaneously identifies an integer address, a direction, a center-to-node radius, an antipodal span, and a complete traversal through the Cell3 center.

8. Integer Plane-Spacing Packets

The reciprocal plane spacings supply an independent route from the measured Q interval to integer Cell3 counts. The adjacent centered-plane distances are the natural spacings for the 27-node half-step scaffold. Dividing Q by each spacing and rounding to the nearest integer gives 101, 74, and 55.

Dimensionally correct Cell3 plane and sheet packets across one measured Q interval

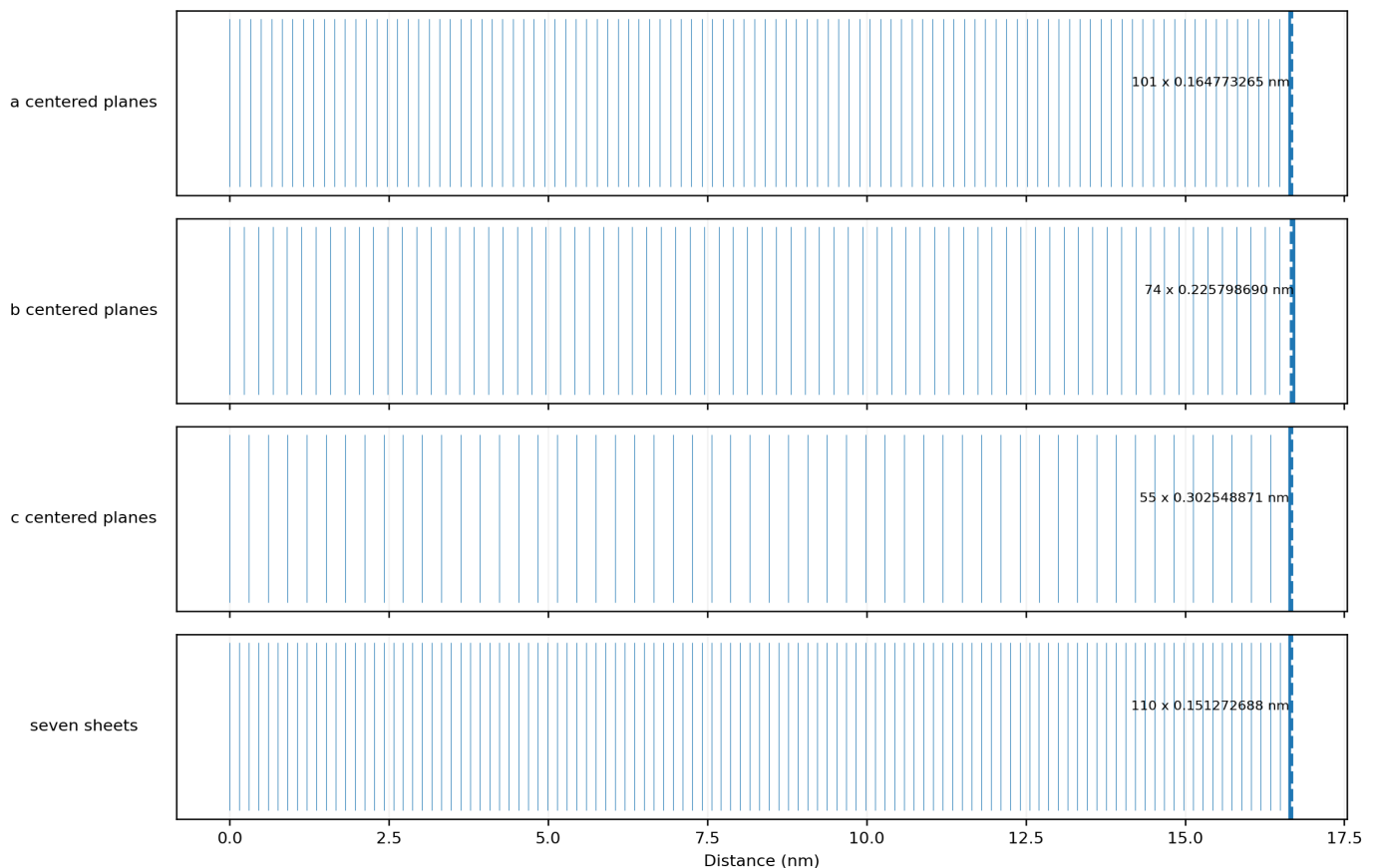


Figure 8. Four dimensionally correct one-dimensional packets across one measured Q interval. Vertical ticks are exact Cell3 centered planes or sheet positions. The solid endpoint is the selected integer packet; the dashed endpoint is measured Q.

Cell3 spacing	Spacing (nm)	Integer count	Packet distance (nm)	Q - packet (nm)	Relative displacement (%)
a centered adjacent-plane spacing	0.164773264855	101	16.642099750313	+0.024566916353	+0.147401
b centered adjacent-plane spacing	0.225798689862	74	16.709103049810	-0.042436383143	-0.254618
c centered adjacent-plane spacing	0.302548870757	55	16.640187891662	+0.026478775004	+0.158873
seven-sheet spacing h	0.151272688042	110	16.639995684672	+0.026670981995	+0.160026
edge length a	0.334179679073	50	16.708983953650	-0.042317286983	-0.253904
edge length c	0.792319765038	21	16.638715065790	+0.027951600877	+0.167710
cross-node center radius r ₁₁₋₁	0.333198543971	50	16.659927198542	+0.006739468124	+0.040437
antipodal cross-node span D ₁₁₋₁	0.666397087942	25	16.659927198542	+0.006739468124	+0.040437

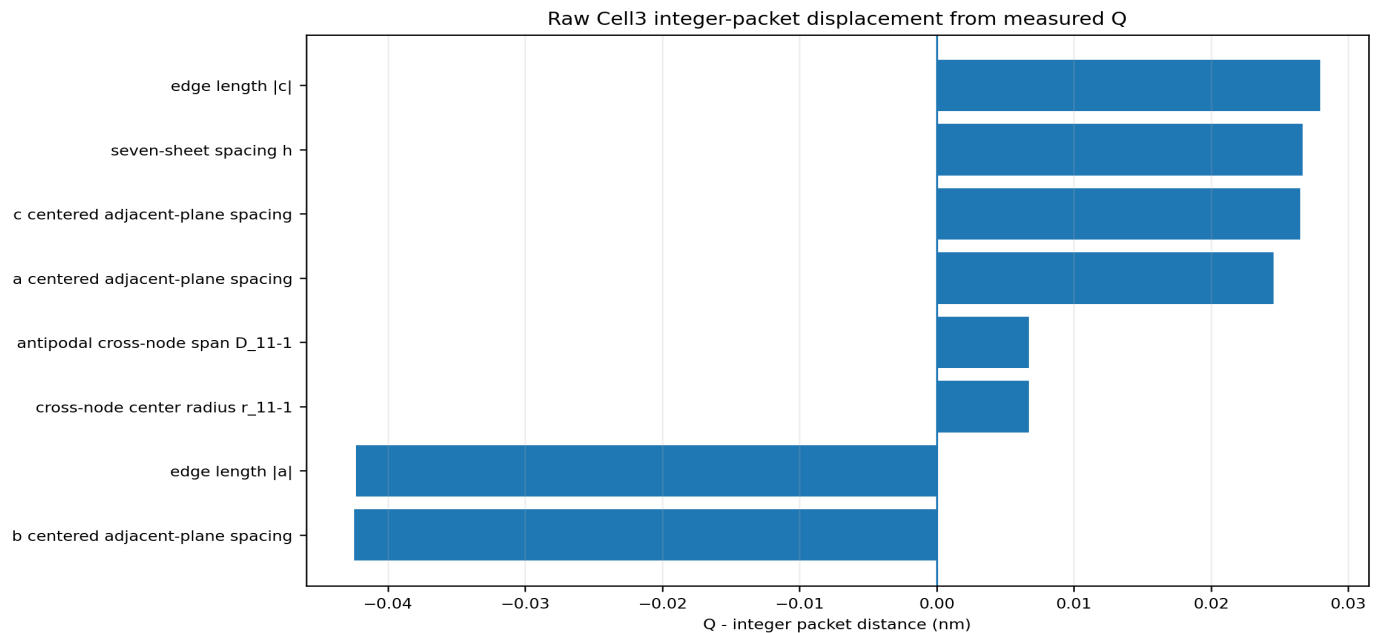


Figure 9. Raw displacement of each integer Cell3 packet from measured Q. Positive values mean the packet ends below Q; negative values mean the packet extends beyond Q.

The same Q interval therefore supports a multi-channel integer description. The packet counts (101,74,55) belong to the three reciprocal coordinate-plane families; 110 belongs to the seven-sheet family; 50 and 25 belong to the direct cross-node radius and antipodal span.

9. Recurrent Spectrophotometer Tracks in Cell3 Addresses

Each recurrent Bragg-plane track has a measured plane orientation and a ladder distance. The quarter-Q representative converts the distance to an exact measured state. Dividing that representative by r_{11-1} converts it to a Cell3 cross-node address count.

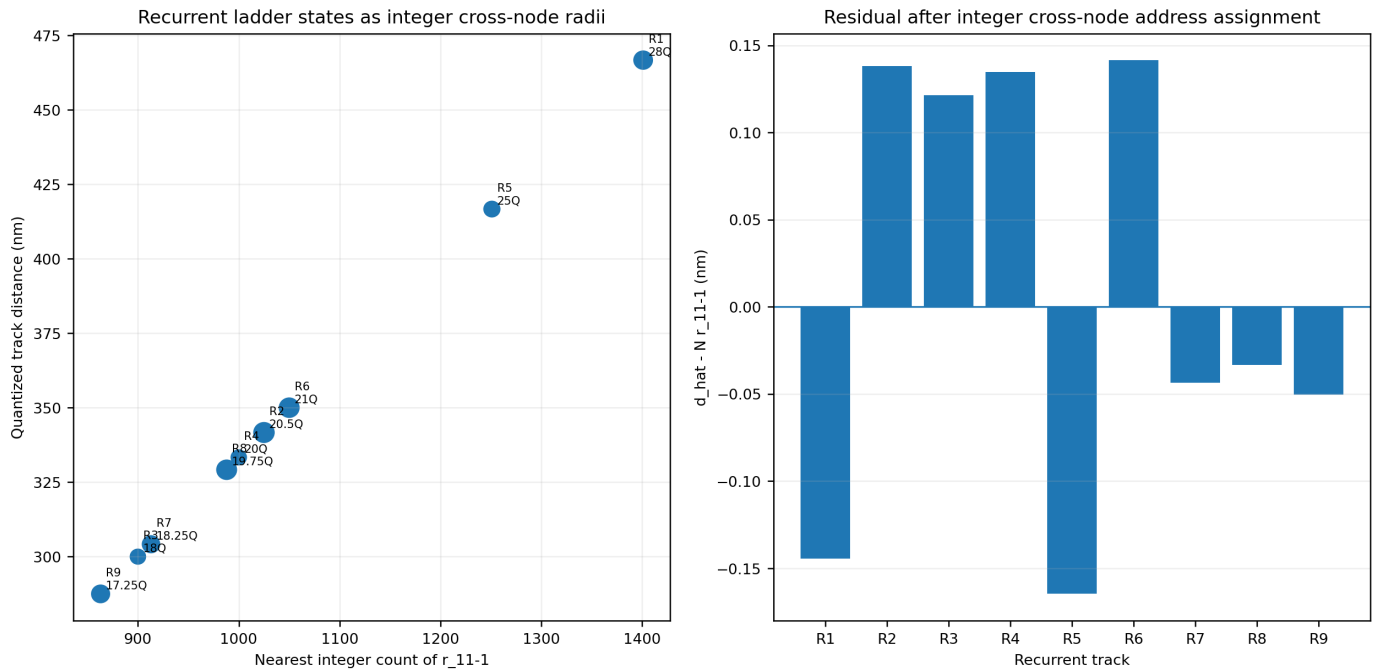


Figure 10. Left: the nine recurrent ladder states plotted at their nearest integer cross-node-radius counts. Right: the residual after assigning each state to that integer count.

Track	Normal (deg)	Raw d (nm)	Q state	d_{hat} (nm)	Nearest r count	r-packet residual (nm)	Nearest D count
R1	7.162697	465.794982	28Q	466.666667	1401	-0.144493	700
R2	31.194545	341.276879	20.5Q	341.666667	1025	+0.138159	513
R3	124.529258	298.281201	18Q	300.000000	900	+0.121310	450
R4	71.229784	331.869612	20Q	333.333333	1000	+0.134789	500
R5	81.201814	416.192497	25Q	416.666667	1251	-0.164712	625
R6	89.598855	350.293683	21Q	350.000000	1050	+0.141529	525
R7	85.818942	304.172812	18.25Q	304.166667	913	-0.043604	456
R8	78.184811	330.523827	19.75Q	329.166667	988	-0.033495	494
R9	98.503824	287.333317	17.25Q	287.500000	863	-0.050343	431

The whole-Q states illustrate the count law directly. One Q interval corresponds to approximately 50 radius units. Consequently 18Q corresponds to approximately 900, 20Q to 1000, 21Q to 1050, 25Q to 1250, and 28Q to 1400. Quarter-Q states add approximately 12.5 radius units before nearest-integer addressing.

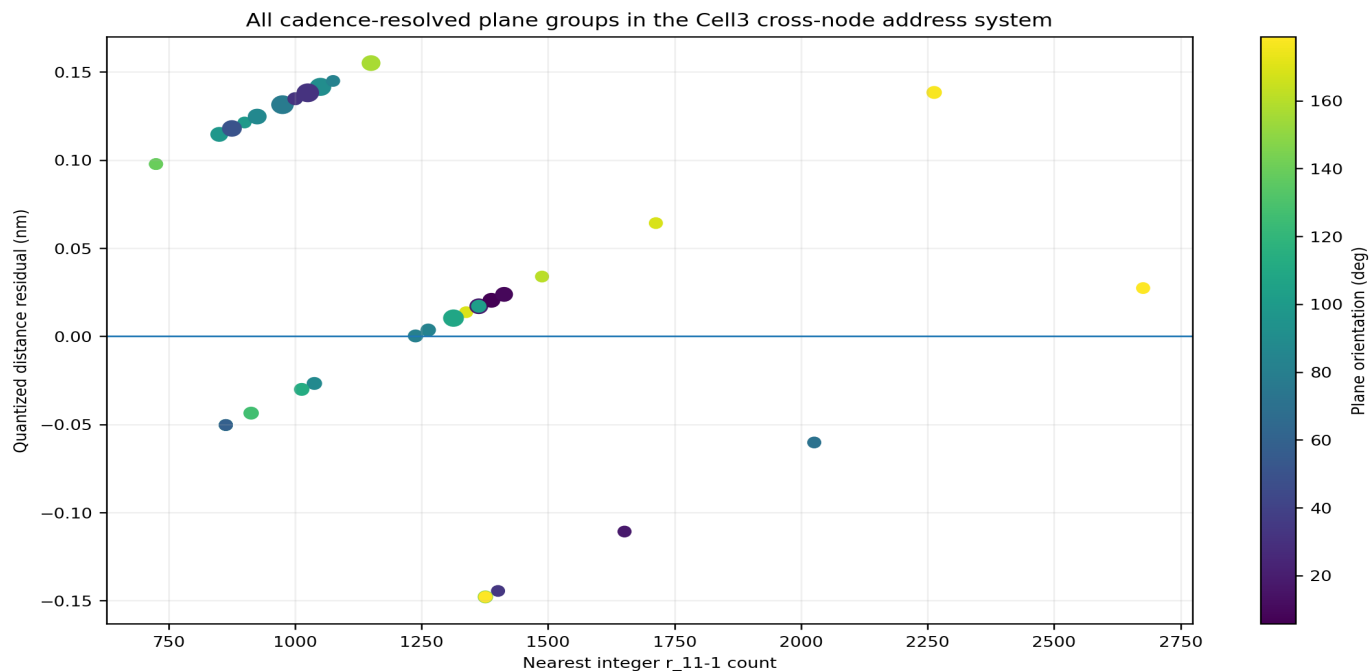


Figure 11. All 41 cadence-resolved plane groups mapped to nearest integer cross-node-radius counts. Point size represents channel-pair support and color represents plane-normal orientation.

10. Plane Orientation and Ladder-State Coupling

A plane state is not fully described by distance alone. Each recurrent track also carries an axial normal angle modulo 180 degrees. The pair (normal, Q level) identifies the projected orientation and measured spacing state; the Cell3 count adds the supercell address.

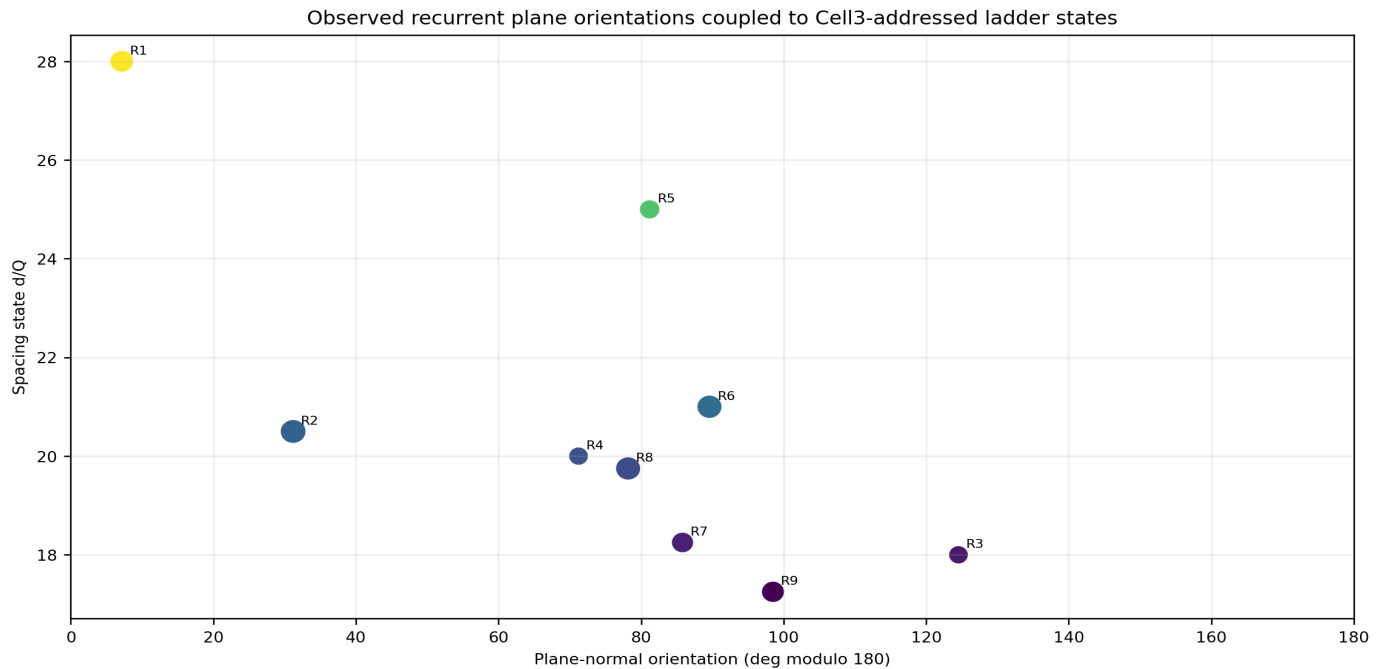


Figure 12. Recurrent plane-normal orientation versus measured Q level. Marker size is total supporting channel-pair count. The near-orthogonal family occupies several distinct Q levels, while the low-angle track near 7.16 degrees occupies 28Q.

The near-orthogonal orientation family around 78-99 degrees occupies 17.25Q, 18.25Q, 19.75Q, 21Q, and 25Q. The 7.16-degree track occupies 28Q. This means orientation selects among several integer supercell packets rather than merely rotating one fixed spacing.

The complete descriptor can be written as (window, normal, k_Q , N_{cross}), where k_Q is the quarter-Q address and N_{cross} is the nearest integer cross-node-radius count. Recurrent tracks compress repeated cadence windows into persistent orientation-distance objects while retaining their source windows and channel support.

11. Supercell Interpretation

A primitive Cell3 is sub-nanometre. The measured Q interval is 16.6666667 nm, about fifty times a characteristic cross-node radius. The recurrent Bragg spacings are hundreds of nanometres and therefore contain hundreds to more than one thousand primitive Cell3 traversal units. The correct geometric interpretation is a supercell or packet scale, not a replacement of the primitive Cell3 metric.

The measured quarter increment $q_d = Q/4$ is likewise not a primitive edge. It is a quarter subdivision of the supercell interval. Its role is to index measured Bragg-plane states. The primitive Cell3 quantities remain $a, b, c, d_a/2, d_b/2, d_c/2, h, r_{11-1}$, and D_{11-1} .

One Q packet can be written in several equivalent approximate integer bases:

Q approximately 50 $r_{11-1} = 25 D_{11-1}$ approximately 101($d_a/2$) approximately 74($d_b/2$) approximately 55($d_c/2$) approximately 110 h

These are not identical geometric directions. They are simultaneous scalar packet counts across different Cell3 structures. The cross-node relation carries a direct node-to-node traversal. The coordinate-plane relations count reciprocal plane crossings. The sheet relation counts affine sheet intervals. Their convergence near one measured Q interval is the central geometric result.

The Cell3 packet therefore acts as a common translation layer between direct-node geometry and reciprocal-plane geometry. This creates a unified address system for measured plane states.

12. Educational Reconstruction

A technical reader can reconstruct the full comparison in five steps.

Step 1 - Build the ordered basis. Place the three authoritative vectors into $B = [a \ b \ c]$. The ICRS basis carries orientation; the local basis carries the same intrinsic geometry in a readable engineering frame.

Step 2 - Build all nodes. For every i, j, k in $\{-1, 0, 1\}$, compute $p = (i \ a + j \ b + k \ c)/2$. Classify a node by the number of zero address coordinates. This produces all 27 nodes.

Step 3 - Build reciprocal spacings. Compute $V = |\det B|$ and the three opposite face areas. Then $d_a = V/A_{bc}$, $d_b = V/A_{ac}$, and $d_c = V/A_{ab}$. Divide by two for adjacent centered planes.

Step 4 - Build the sheet and cross-node objects. Solve $B^T N$ proportional to $(1, 1, 1)$ for the sheet normal and spacing h . Compute $(a+b-c)/2$ for the selected cross-node radius.

Step 5 - Compare the measured Q . Divide Q by each Cell3 spacing, assign the nearest integer packet count, and retain the endpoint residual. Apply the same procedure to every recurrent ladder state.

This procedure retains both data streams without conflation: the spectrophotometer supplies Q , q_d , plane orientations, and recurrent distances; the Cell3 geometry supplies primitive vectors, nodes, plane families, sheets, and cross-node spacings.

13. New Pathways to Discovery

The comparison opens a direct experimental route from broadband optical phase measurements to integer Cell3 addresses. A measured interplanar distance can now be represented simultaneously by a Bragg-plane normal, a quarter-Q state, a coordinate-plane packet count, a sheet count, and a cross-node traversal count.

The first pathway is cadence-resolved address tracking. Repeating the 270-wide, 90-cadence analysis over longer records will produce time-ordered sequences of Cell3 packet addresses rather than isolated floating-point distances.

The second pathway is multi-orientation closure. The present data show several plane orientations at distinct Q levels. A larger orientation set can be compared with the full direct and reciprocal Cell3 direction families to identify complete supercell plane systems.

The third pathway is cross-channel reconstruction. Each optical channel pair contributes a Bragg solution. Grouping channel-pair solutions by shared Cell3 packet addresses provides a direct method for separating plane families and cross-node families.

The fourth pathway is device-scale geometry. Once an integer packet is identified, a physical device trajectory can be expressed as a declared number of primitive plane crossings, sheet crossings, or antipodal node spans. This replaces a dimensionless fit with a geometric count.

The immediate archival conclusion is that the spectrophotometer ladder approximates Cell3 integer planar spacings and cross-node spacings. The observed Q interval is the organizing supercell packet, and the (1,1,-1) cross-node direction supplies the most direct integer closure.

Appendix A. Equation Ledger

Object	Equation
Cell3 basis	$B = [a \ b \ c]$
Centered node	$p_L = (1/2) B L, L \text{ in } \{-1,0,1\}^3$
Direct metric	$M = B^T B$
Reciprocal metric	$R = M^{-1}$
Cell volume	$V = \det B $
Plane spacings	$d_a = V/A_{bc}; d_b = V/A_{ac}; d_c = V/A_{ab}$
Centered plane spacings	$s_a = d_a/2; s_b = d_b/2; s_c = d_c/2$
Sheet index	$s = i + j + k$
Sheet coordinate	$N_{\text{sheet}} \cdot p_L = s \ h$
Cross-node radius	$r_{11-1} = a+b-c /2$
Antipodal span	$D_{11-1} = a+b-c $
Measured ladder	$Q = 50/3 \text{ nm}; q_d = Q/4$
Measured state index	$k = \text{round}(d/q_d)$
Measured representative	$d_{\text{hat}} = k \ q_d$
Cell3 packet count	$N_x = \text{round}(Q/x) \text{ or } \text{round}(d_{\text{hat}}/x)$
Packet residual	$\epsilon_x = \text{target} - N_x \ x$

Appendix B. Canonical Cell3 Metric Ledger

Quantity	Value	Unit / meaning
det(B)	-0.117914891916764	nm ³ , left-handed local basis
V	0.117914891916764	nm ³
A_ab	0.194869165469925	nm ²
A_ac	0.261106235799411	nm ²
A_bc	0.357809539128890	nm ²
d_a	0.329546529709173	nm
d_b	0.451597379724587	nm
d_c	0.605097741515000	nm
d_a/2	0.164773264854586	nm
d_b/2	0.225798689862294	nm
d_c/2	0.302548870757500	nm
h	0.151272688042469	nm
r_11-1	0.333198543970849	nm
D_11-1	0.666397087941697	nm

Direct metric M

	a	b	c
a	0.111676057905333	0.020710333281645	0.043937235097629
b	0.020710333281645	0.343877732397534	0.296392758961495
c	0.043937235097629	0.296392758961495	0.627770610069268

Reciprocal metric R

	a*	b*	c*
a*	9.208025528690507	0.001534802423075	-0.645188035672761
b*	0.001534802423075	4.903398312296265	-2.315175584411565
c*	-0.645188035672761	-2.315175584411565	2.731171274743789

Appendix C. Complete 27-Node Cell3 Ledger

All positions use the authoritative local engineering basis and are stated in nanometres.

Address	Class	x (nm)	y (nm)	z (nm)	Radius (nm)	Sheet s	Sheet coord. (nm)
(-1,-1,-1)	corner	-0.263815601795	-0.538717711623	0.302548870757	0.671826810840	-3	-0.453818064127
(-1,-1,0)	edge center	-0.198076662761	-0.291563457734	-0.000000000000	0.352482076447	-2	-0.302545376085
(-1,-1,1)	corner	-0.132337723728	-0.044409203845	-0.302548870757	0.333198543971	-1	-0.151272688042
(-1,0,-1)	edge center	-0.232828778570	-0.247154253889	0.302548870757	0.454785976633	-2	-0.302545376085
(-1,0,0)	face center	-0.167089839537	0.000000000000	0.000000000000	0.167089839537	-1	-0.151272688042
(-1,0,1)	edge center	-0.101350900503	0.247154253889	-0.302548870757	0.403600110809	0	0.000000000000
(-1,1,-1)	corner	-0.201841955345	0.044409203845	0.302548870757	0.366398924016	-1	-0.151272688042
(-1,1,0)	edge center	-0.136103016312	0.291563457734	0.000000000000	0.321765879072	0	0.000000000000
(-1,1,1)	corner	-0.070364077278	0.538717711623	-0.302548870757	0.621855043707	1	0.151272688042
(0,-1,-1)	edge center	-0.096725762259	-0.538717711623	0.302548870757	0.625386652478	-2	-0.302545376085
(0,-1,0)	face center	-0.030986823225	-0.291563457734	0.000000000000	0.293205445208	-1	-0.151272688042
(0,-1,1)	edge center	0.034752115809	-0.044409203845	-0.302548870757	0.307759169053	0	0.000000000000
(0,0,-1)	face center	-0.065738939034	-0.247154253889	0.302548870757	0.396159882519	-1	-0.151272688042
(0,0,0)	body center	0.000000000000	0.000000000000	0.000000000000	0.000000000000	0	0.000000000000
(0,0,1)	face center	0.065738939034	0.247154253889	-0.302548870757	0.396159882519	1	0.151272688042
(0,1,-1)	edge center	-0.034752115809	0.044409203845	0.302548870757	0.307759169053	0	0.000000000000
(0,1,0)	face center	0.030986823225	0.291563457734	0.000000000000	0.293205445208	1	0.151272688042
(0,1,1)	edge center	0.096725762259	0.538717711623	-0.302548870757	0.625386652478	2	0.302545376085
(1,-1,-1)	corner	0.070364077278	-0.538717711623	0.302548870757	0.621855043707	-1	-0.151272688042
(1,-1,0)	edge center	0.136103016312	-0.291563457734	0.000000000000	0.321765879072	0	0.000000000000
(1,-1,1)	corner	0.201841955345	-0.044409203845	-0.302548870757	0.366398924016	1	0.151272688042
(1,0,-1)	edge center	0.101350900503	-0.247154253889	0.302548870757	0.403600110809	0	0.000000000000
(1,0,0)	face center	0.167089839537	0.000000000000	0.000000000000	0.167089839537	1	0.151272688042
(1,0,1)	edge center	0.232828778570	0.247154253889	-0.302548870757	0.454785976633	2	0.302545376085
(1,1,-1)	corner	0.132337723728	0.044409203845	0.302548870757	0.333198543971	1	0.151272688042
(1,1,0)	edge center	0.198076662761	0.291563457734	0.000000000000	0.352482076447	2	0.302545376085
(1,1,1)	corner	0.263815601795	0.538717711623	-0.302548870757	0.671826810840	3	0.453818064127

Appendix D. Coordinate-Plane Membership Ledger

Family	Address value	Plane coordinate (nm)	Adjacent spacing (nm)	Unit normal	Nine node addresses
a	-1	-0.164773264855	0.164773264855	(-0.98613575,+0.10480468,-0.12865557)	(-1,-1,-1); (-1,-1,0); (-1,-1,1); (-1,0,-1); (-1,0,0); (-1,0,1); (-1,1,-1); (-1,1,0); (-1,1,1)
a	0	0.000000000000	0.164773264855	(-0.98613575,+0.10480468,-0.12865557)	(0,-1,-1); (0,-1,0); (0,-1,1); (0,0,-1); (0,0,0); (0,0,1); (0,1,-1); (0,1,0); (0,1,1)
a	1	0.164773264855	0.164773264855	(-0.98613575,+0.10480468,-0.12865557)	(1,-1,-1); (1,-1,0); (1,-1,1); (1,0,-1); (1,0,0); (1,0,1); (1,1,-1); (1,1,0); (1,1,1)
b	-1	-0.225798689862	0.225798689862	(+0.00000000,-0.77444098,-0.63264616)	(-1,-1,-1); (-1,-1,0); (-1,-1,1); (0,-1,-1); (0,-1,0); (0,-1,1); (1,-1,-1); (1,-1,0); (1,-1,1)
b	0	0.000000000000	0.225798689862	(+0.00000000,-0.77444098,-0.63264616)	(-1,0,-1); (-1,0,0); (-1,0,1); (0,0,-1); (0,0,0); (0,0,1); (1,0,-1); (1,0,0); (1,0,1)
b	1	0.225798689862	0.225798689862	(+0.00000000,-0.77444098,-0.63264616)	(-1,1,-1); (-1,1,0); (-1,1,1); (0,1,-1); (0,1,0); (0,1,1); (1,1,-1); (1,1,0); (1,1,1)
c	-1	-0.302548870757	0.302548870757	(+0.00000000,+0.00000000,+1.00000000)	(-1,-1,-1); (-1,0,-1); (-1,1,-1); (0,-1,-1); (0,0,-1); (0,1,-1); (1,-1,-1); (1,0,-1); (1,1,-1)
c	0	0.000000000000	0.302548870757	(+0.00000000,+0.00000000,+1.00000000)	(-1,-1,0); (-1,0,0); (-1,1,0); (0,-1,0); (0,0,0); (0,1,0); (1,-1,0); (1,0,0); (1,1,0)
c	1	0.302548870757	0.302548870757	(+0.00000000,+0.00000000,+1.00000000)	(-1,-1,1); (-1,0,1); (-1,1,1); (0,-1,1); (0,0,1); (0,1,1); (1,-1,1); (1,0,1); (1,1,1)

Appendix E. Complete Recurrent Track Cell3 Mapping

Track	Normal	Windows	Support	Raw d	Q level	d_hat	r count	r residual	D count	a planes	b planes	c planes	sheets
R1	7.162697	6, 7	10	465.794982	28	466.666667	1401	-0.144493	700	2832	2067	1542	3085
R2	31.194545	8, 9	13	341.276879	20.5	341.666667	1025	+0.138159	513	2074	1513	1129	2259
R3	124.529258	3, 9	5	298.281201	18	300.000000	900	+0.121310	450	1821	1329	992	1983
R4	71.229784	4, 7	5	331.869612	20	333.333333	1000	+0.134789	500	2023	1476	1102	2204
R5	81.201814	1, 3	6	416.192497	25	416.666667	1251	-0.164712	625	2529	1845	1377	2754
R6	89.598855	3, 4	12	350.293683	21	350.000000	1050	+0.141529	525	2124	1550	1157	2314
R7	85.818942	5, 7	8	304.172812	18.25	304.166667	913	-0.043604	456	1846	1347	1005	2011
R8	78.184811	6, 9	12	330.523827	19.75	329.166667	988	-0.033495	494	1998	1458	1088	2176
R9	98.503824	1, 2, 4	9	287.333317	17.25	287.500000	863	-0.050343	431	1745	1273	950	1901

Appendix F. Complete 41-Group Quantization and Cross-Node Address Ledger

Group	Win	Support	Normal	Raw d	Quarter k	Q level	d_hat	Q residual	Cross count	Cross residual
W1P1	1	6	156.018218	382.179999	92	23	383.333333	-1.153335	1150	+0.155008
W1P2	1	2	127.494751	241.959760	58	14.5	241.666667	+0.293093	725	+0.097722
W1P3	1	2	142.887062	292.266313	70	17.5	291.666667	+0.599646	875	+0.117941
W1P4	1	2	168.299955	571.720493	137	34.25	570.833333	+0.887160	1713	+0.064228
W1P5	1	2	101.011966	288.198533	69	17.25	287.500000	+0.698533	863	-0.050343
W1P6	1	3	80.620654	413.191276	99	24.75	412.500000	+0.691276	1238	+0.000203
W2P1	2	2	139.301436	241.109920	58	14.5	241.666667	-0.556747	725	+0.097722
W2P2	2	2	169.410007	446.776170	107	26.75	445.833333	+0.942837	1338	+0.013682
W2P3	2	2	111.264847	287.680097	69	17.25	287.500000	+0.180097	863	-0.050343
W2P4	2	2	98.871992	298.282048	72	18	300.000000	-1.717952	900	+0.121310
W3P1	3	9	90.326728	351.212701	84	21	350.000000	+1.212701	1050	+0.141529
W3P2	3	3	81.782975	419.215517	101	25.25	420.833333	-1.617816	1263	+0.003572
W3P3	3	2	70.741052	675.889591	162	40.5	675.000000	+0.889591	2026	-0.060250
W3P4	3	2	121.934095	287.479519	69	17.25	287.500000	-0.020481	863	-0.050343
W4P1	4	3	72.960011	331.457689	80	20	333.333333	-1.875644	1000	+0.134789
W4P2	4	5	97.354278	282.726960	68	17	283.333333	-0.606374	850	+0.114571
W4P3	4	2	17.627769	549.550602	132	33	550.000000	-0.449398	1651	-0.110796
W4P4	4	3	87.413352	347.551033	83	20.75	345.833333	+1.717700	1038	-0.026755
W4P5	4	4	71.041940	460.670327	111	27.75	462.500000	-1.829673	1388	+0.020421
W4P6	4	3	112.137136	337.901028	81	20.25	337.500000	+0.401028	1013	-0.030125
W4P7	4	2	178.956728	892.592963	214	53.5	891.666667	+0.926296	2676	+0.027363
W5P1	5	6	86.401611	309.357186	74	18.5	308.333333	+1.023853	925	+0.124680
W5P2	5	6	18.626817	453.039276	109	27.25	454.166667	-1.127390	1363	+0.017051
W5P3	5	2	33.469916	467.896181	112	28	466.666667	+1.229515	1401	-0.144493
W6P1	6	8	108.014052	436.630398	105	26.25	437.500000	-0.869602	1313	+0.010312
W6P2	6	5	5.804736	462.016971	111	27.75	462.500000	-0.483029	1388	+0.020421
W6P3	6	2	81.671158	357.719932	86	21.5	358.333333	-0.613402	1075	+0.144899
W7P1	7	2	84.069968	289.135186	69	17.25	287.500000	+1.635186	863	-0.050343
W7P2	7	3	102.273489	454.693382	109	27.25	454.166667	+0.526715	1363	+0.017051
W7P3	7	3	177.434914	755.615584	181	45.25	754.166667	+1.448918	2263	+0.138362
W7P4	7	5	8.520658	469.603887	113	28.25	470.833333	-1.229447	1413	+0.023791
W7P5	7	2	58.527808	288.445371	69	17.25	287.500000	+0.945371	863	-0.050343
W7P6	7	2	68.632463	332.488455	80	20	333.333333	-0.844879	1000	+0.134789
W8P1	8	3	145.821385	456.955884	110	27.5	458.333333	-1.377449	1376	-0.147863
W8P2	8	7	50.295361	291.791273	70	17.5	291.666667	+0.124606	875	+0.117941
W8P3	8	10	31.263378	343.419091	82	20.5	341.666667	+1.752425	1025	+0.138159
W9P1	9	10	77.489193	325.337931	78	19.5	325.000000	+0.337931	975	+0.131420
W9P2	9	3	126.258051	305.706869	73	18.25	304.166667	+1.540203	913	-0.043604
W9P3	9	3	30.965096	334.232212	80	20	333.333333	+0.898878	1000	+0.134789
W9P4	9	2	160.744652	495.847059	119	29.75	495.833333	+0.013725	1488	+0.033900
W9P5	9	2	178.458100	457.644538	110	27.5	458.333333	-0.688795	1376	-0.147863

Appendix G. Source Integrity and Archival Record

Source file	SHA-256
K-Field_spectrophotometer_quantized_interplanar_distance_ladder_1785505620.json	d9dd8322a7cf2bbc8c311201d5785d3ba82bc09ac677a64a38776118859eda61
K-Field_T54p17_270x90_plane_groups_1785504180.csv	38ca5818818a4185343c2c9675686a314b34381139b352f9191024af7be2ad98
K-Field_T54p17_270x90_recurrent_plane_tracks_1785504180.csv	d56619ba6abaa7e0813a7434c49be84e1d3a310262882f9c57d2cc177681301f
K-Field_T54p17_270x90_phase_by_window_1785504180.csv	0d0be00f607e2b8088e703900de8016a119f09c0fbb9cbce697bf274d12e83e7
K-Field_T54p17_270x90_bragg_solutions_1785504180.csv	d0d43258865e0d04ff782594fe584a9d6ca760241d4b6a84a84d03ccec96b9cf
spectrometer_test_log(1).txt	681bef37087dad25ebc71c605e0adc00f95a2b1d3b9415594a16bc109ee655fb
K-Field_Converged_Closure_1783815468(3).pdf	ad381e1164cbd14bc2e388180aa46a9fcbc72ce3ad959b5aecc7379e767d51a9
K-Field_lab_scale_device_proposals_1784994055.pdf	237d3f1eb817d8b246ad51096850a601e42bdcf0df1259bb4e86cd15f2eb5dde
K-Field_spectrophotometer_quantized_interplanar_distance_ladder_1785505620.pdf	36f4d6c1307a65058b3e4a6c97095b7bdc1faa5cb27a5d96e35dc1683d5619b5

Field	Value
Principal Investigator	P. Dwayne Esterline
Degrees	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx, LLC.
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation epoch	1785507240
Generation UTC	2026-07-31T14:14:00Z
PDF	K-Field_cell3_vs_spectrophotometer_quantized_interplanar_distance_ladder_1785507240.pdf
JSON	K-Field_cell3_vs_spectrophotometer_quantized_interplanar_distance_ladder_1785507240.json
PDF engine	Python ReportLab
PDF fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier only
Historical format references	K-Field Spectrophotometer Quantized Interplanar Distance Ladder; K-Field Spectrophotometer Optimized Geometry; K-Field Converged Closure; K-Field Lab-Scale Device Proposals
Numerical basis	Measured 270x90 Bragg-plane ledgers and authoritative Cell3 nanometre basis
Machine-readable scope	Complete basis, metrics, 27 nodes, coordinate planes, seven sheets, cross-node objects, integer packets, recurrent tracks, 41 plane groups, figures, hashes, and generation metadata

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